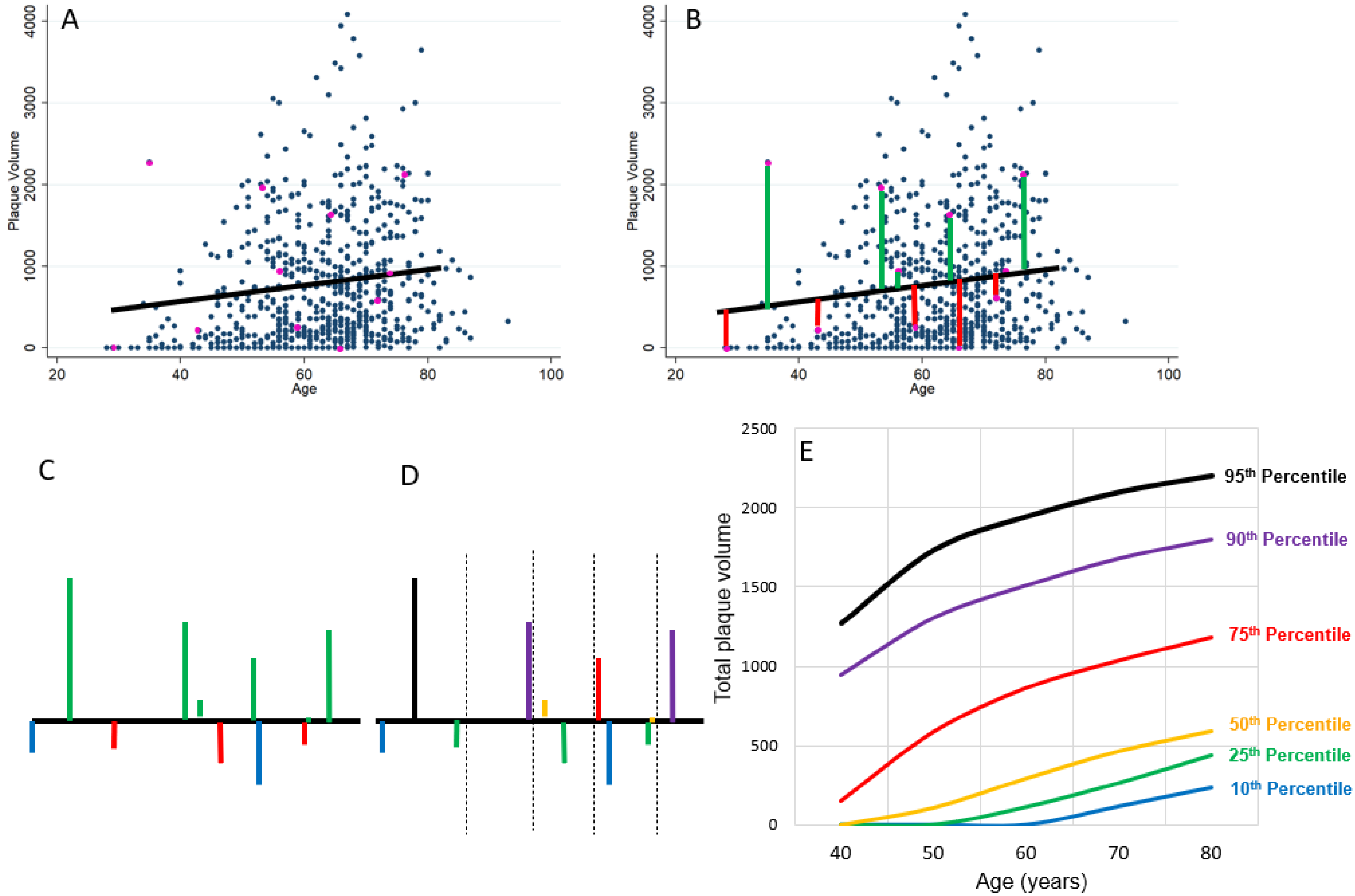
SUPPLEMENTAL MATERIAL

Supplemental Methods

DL Architecture

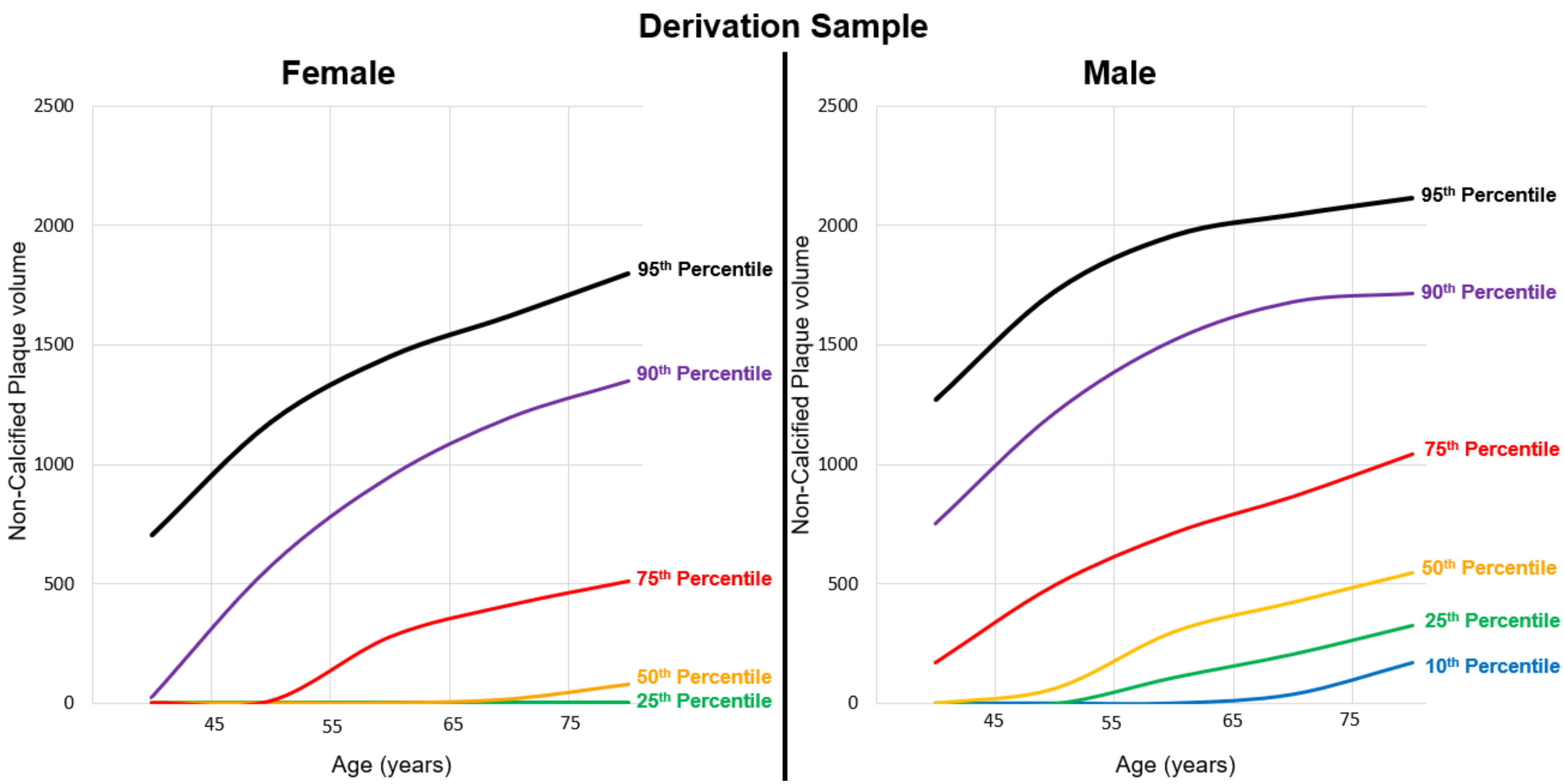
CTA vessel cross-sections with a 0.3 mm slice thickness and 20 mm field of view around predefined coronary centerlines are used as input data for the convolutional long short-term memory (ConvLSTM) network. Centerlines generated through standard CTA reporting software can be utilized for this purpose. The hierarchical ConvLSTM network has two branches. The first branch uses a ConvLSTM to extract features from the current cross-section and five adjacent sections on each side(14). The second uses a DenseNet block to extract features from the current cross-section only(14). The outputs of both branches are combined using an attention head to generate a probability of each voxel belonging to a specific label. In our previous work the mean computation speed for our network was 0.16 seconds with a graphics processing unit and 4.3 seconds with central processing units per patient(16). In our previous work the mean computation speed for our network was 0.16 seconds with a graphics processing unit and 4.3 seconds with central processing units per patient(14). Physicians or trained technical staff then evaluate the plaque segmentation and correct contours as required. The total time required for this quality control process typically ranges from 5-10 minutes depending on case complexity. The Dice score for lumen and plaque segmentation compared to manual expert segmentation was 0.90 (14).

Figure S1



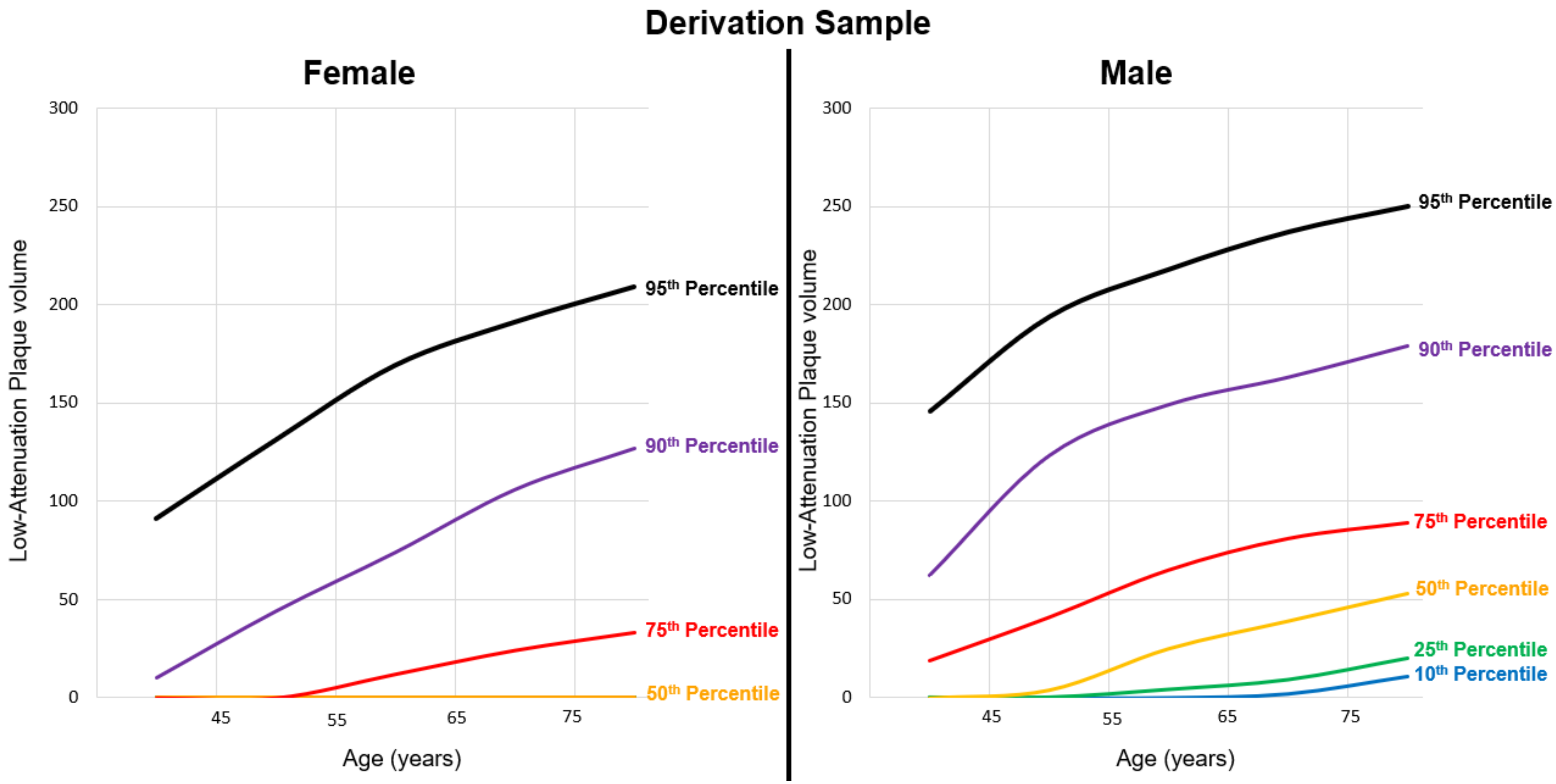
Overview of the process for determining population percentiles. Panel A: Representative scatterplot of age and total plaque volume, with selected patients in magenta. A random-effects regression model was developed to predict coronary plaque volumes (black line). In B, differences between actual plaque volumes and expected plaque volumes were determined (more plaque volume than expected in green, less than expected in red). In panel C, differences are assigned a raw percentile (with zero plaque volume as dark blue). Percentiles are then re-categorized to be a function of non-zero values in age and sex-specific categories (panel D). These final percentiles were utilized to establish expected population distributions (panel E).

Figure S2



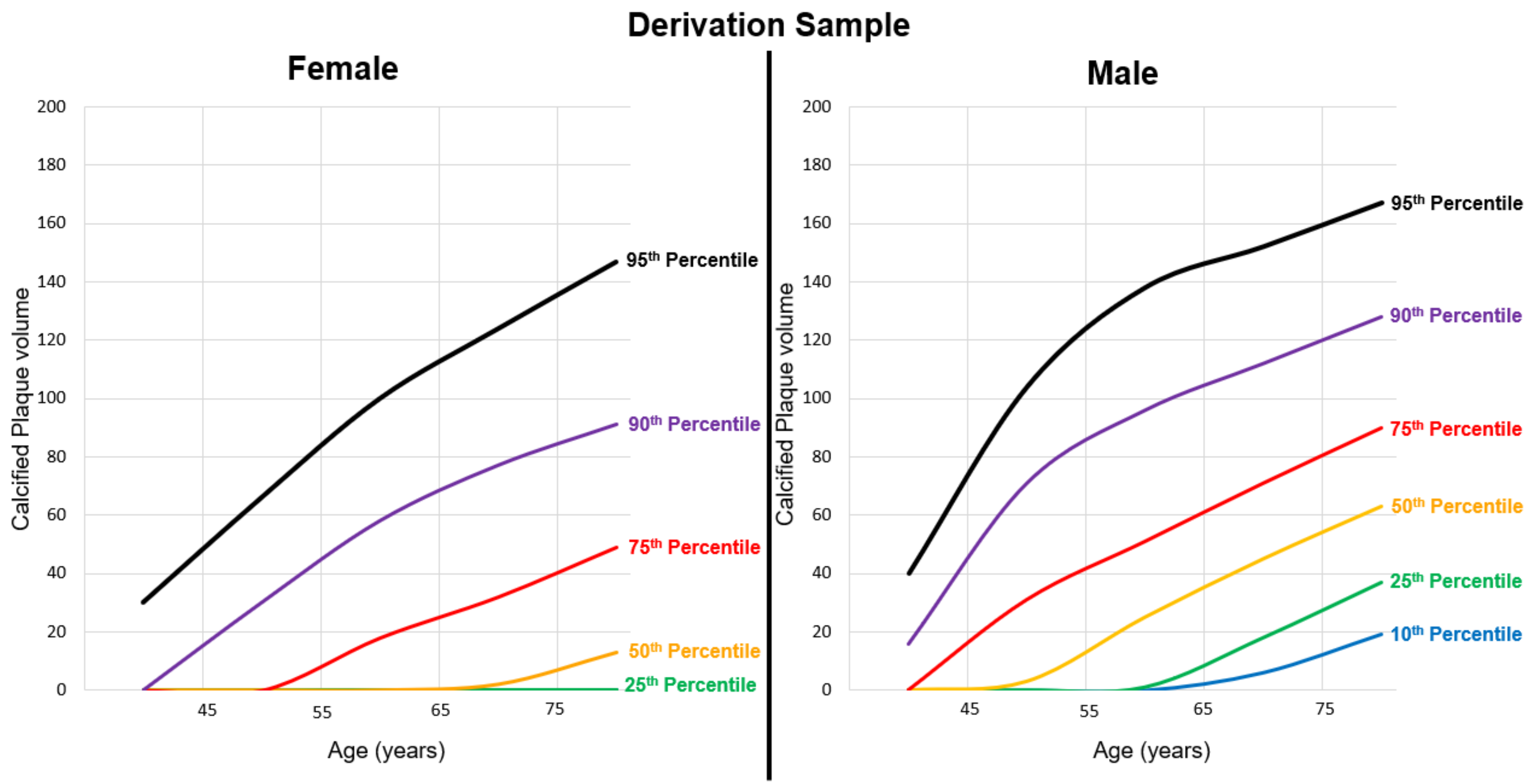
Distribution of non-calcified plaque volume as a function of sex and age. Expected plaque volumes increase with age in female and male patients, with overall higher expected plaque volumes in male patients.

Figure S3



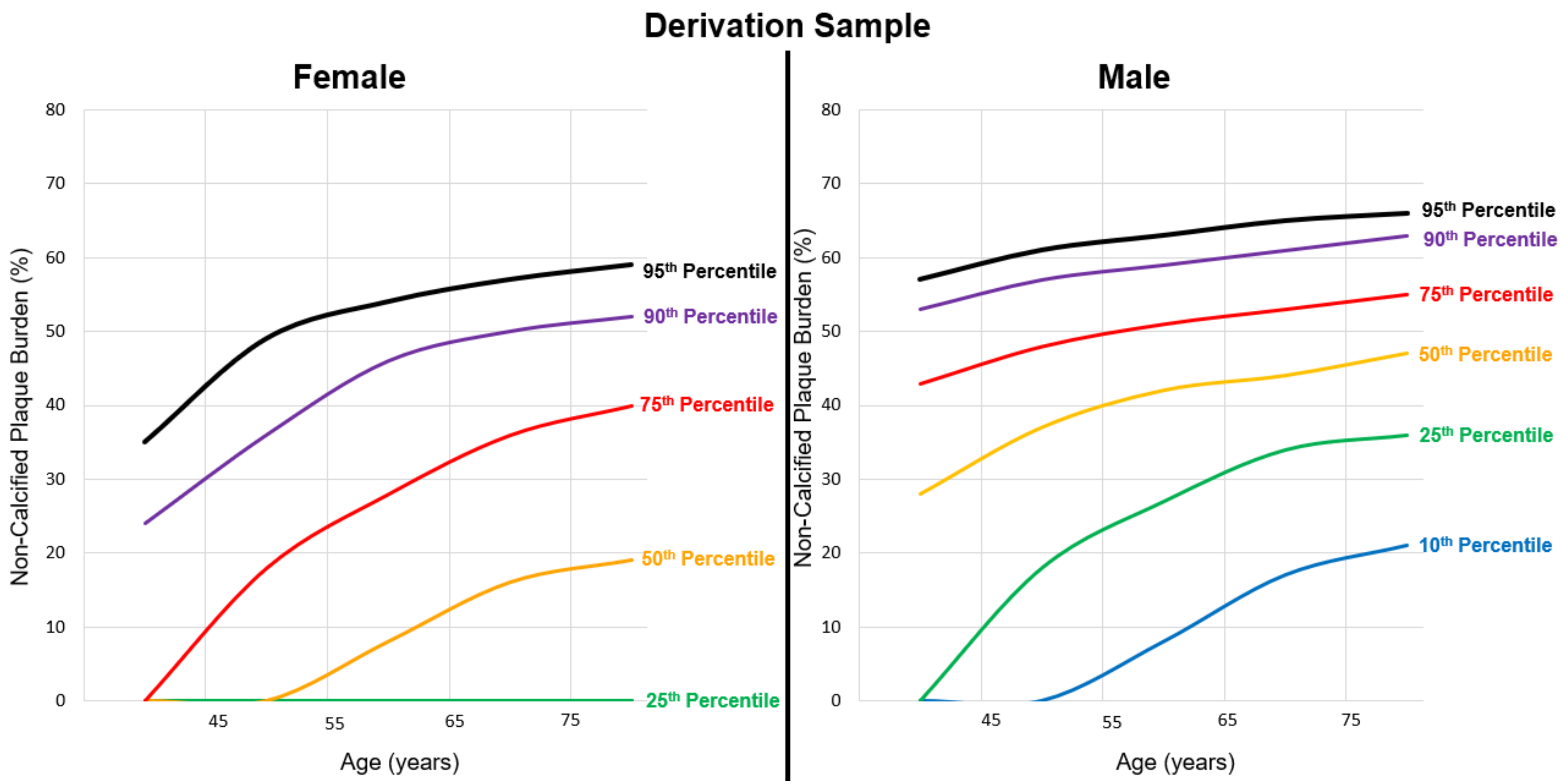
Distribution of low-attenuation plaque volume as a function of sex and age. Expected plaque volumes increase with age in female and male patients, with overall higher expected plaque volumes in male patients. The 50th percentile (orange) for female patients remained at $0mm^3$ for all age groups.

Figure S4



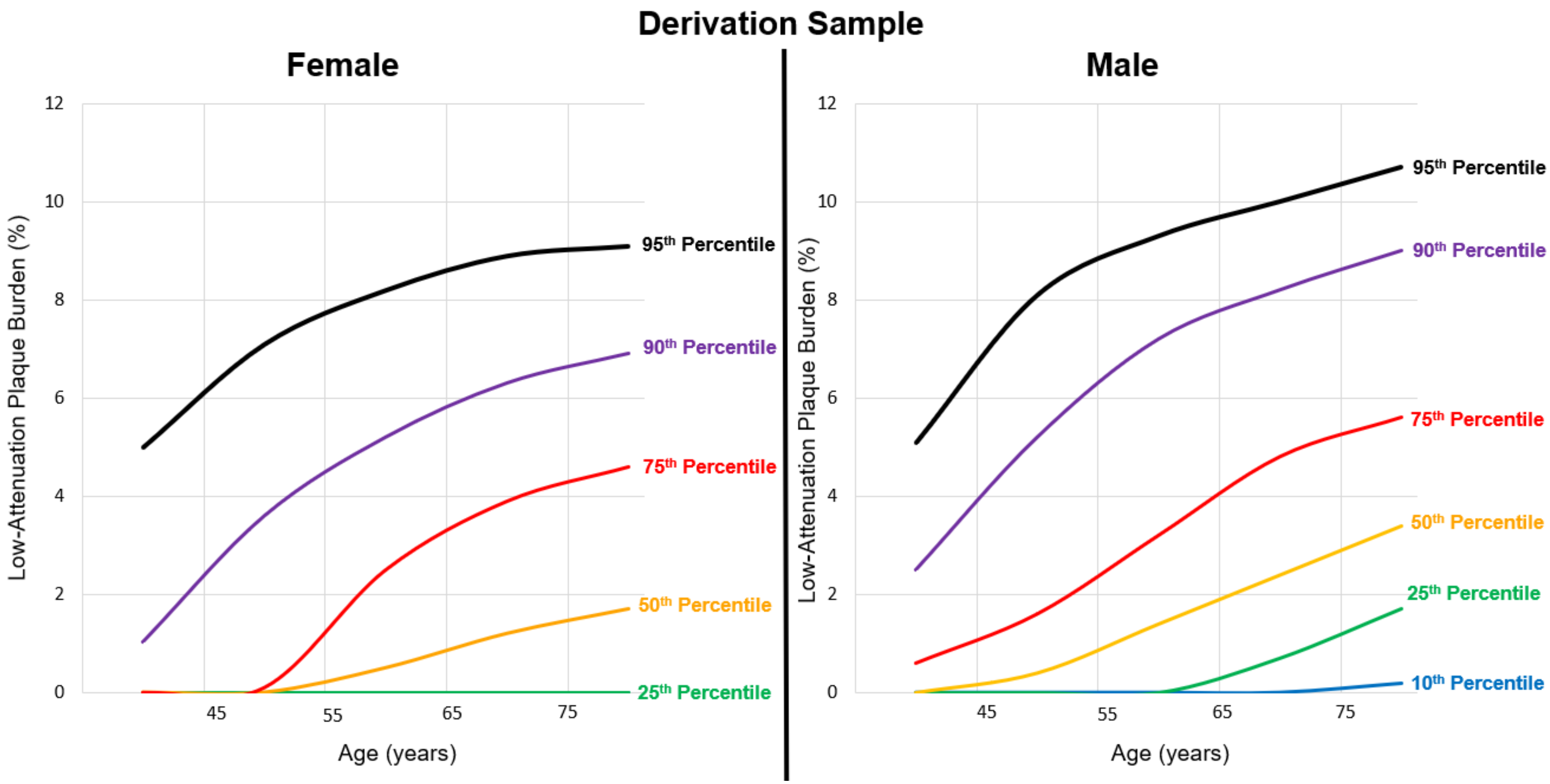
Distribution of calcified plaque volume as a function of sex and age. Expected plaque volume increase with age in female and male patients, with overall higher expected plaque volumes in male patients.

Figure S5



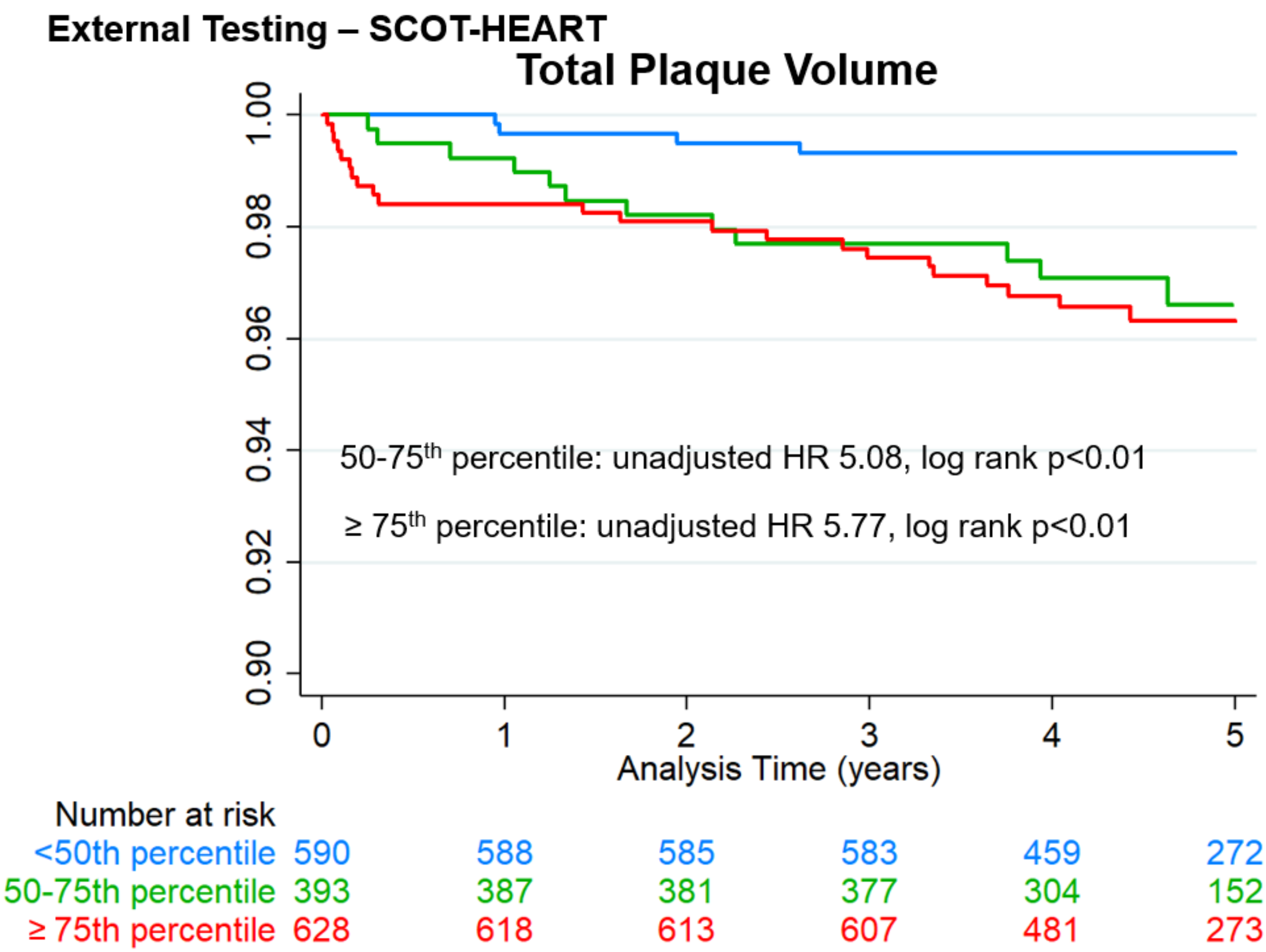
Distribution of non-calcified plaque burden as a function of sex and age. Plaque burden is calculated as plaque volume/analyzed segment volume Expected plaque burden increase with age in female and male patients, with overall higher expected plaque volumes in male patients.

Figure S6



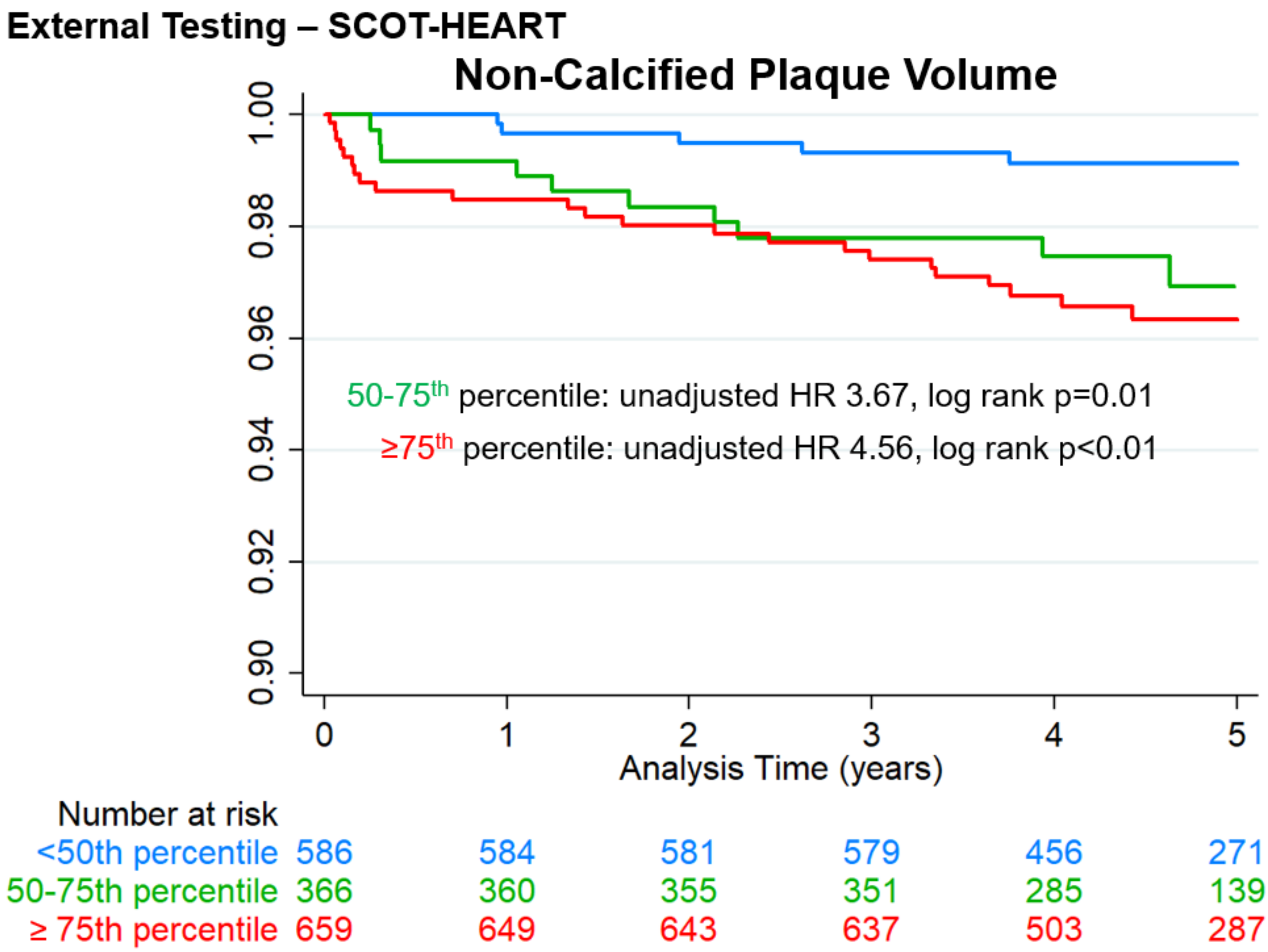
Distribution of low-attenuation plaque burden as a function of sex and age. Plaque burden is calculated as plaque volume/analyzed segment volume. Expected plaque burden increase with age in female and male patients, with overall higher expected plaque volumes in male patients.

Figure S7



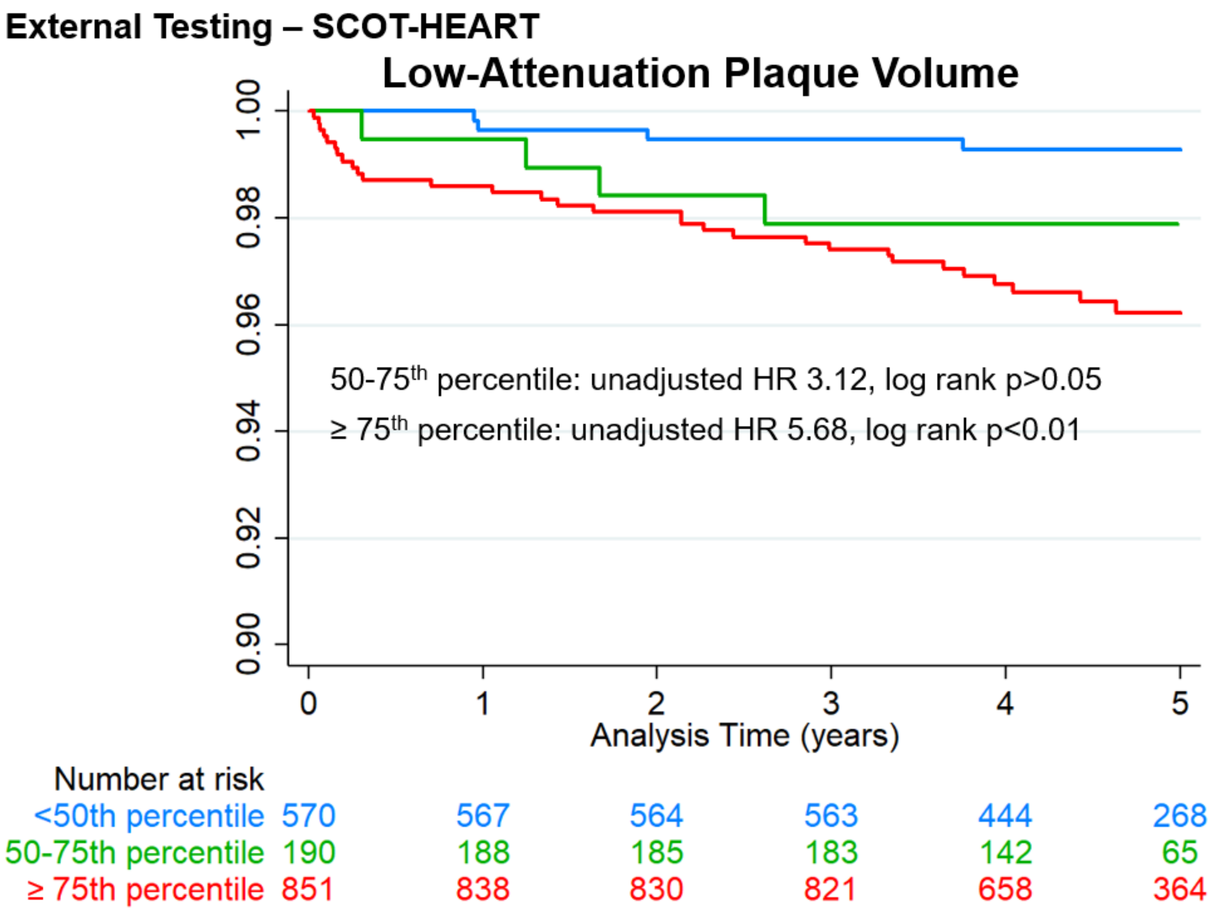
Survival free of fatal or non-fatal myocardial infarction stratified by percentile of total plaque volume. Reference population was patients in the <50th percentile for age and sex. The percentiles were derived in a separate population, then applied to a cohort of patients from the Scottish Computed Tomography of the Heart trial. HR – hazard ratio

Figure S8



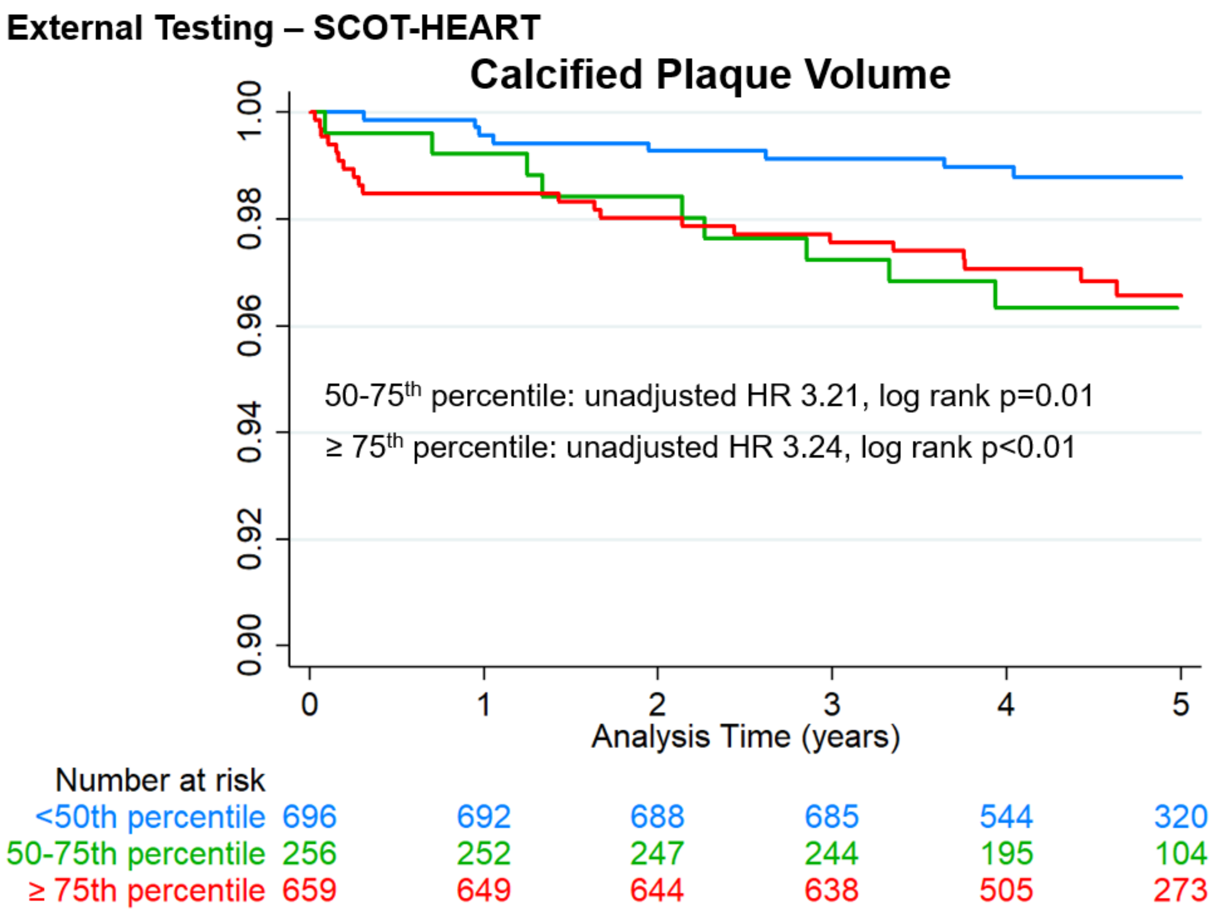
Survival free of fatal or non-fatal myocardial infarction stratified by percentile of non-calcified plaque volume. Reference population was patients in the <50th percentile for age and sex. The percentiles were derived in a separate population, then applied to a cohort of patients from the Scottish Computed Tomography of the Heart trial. HR – hazard ratio

Figure S9



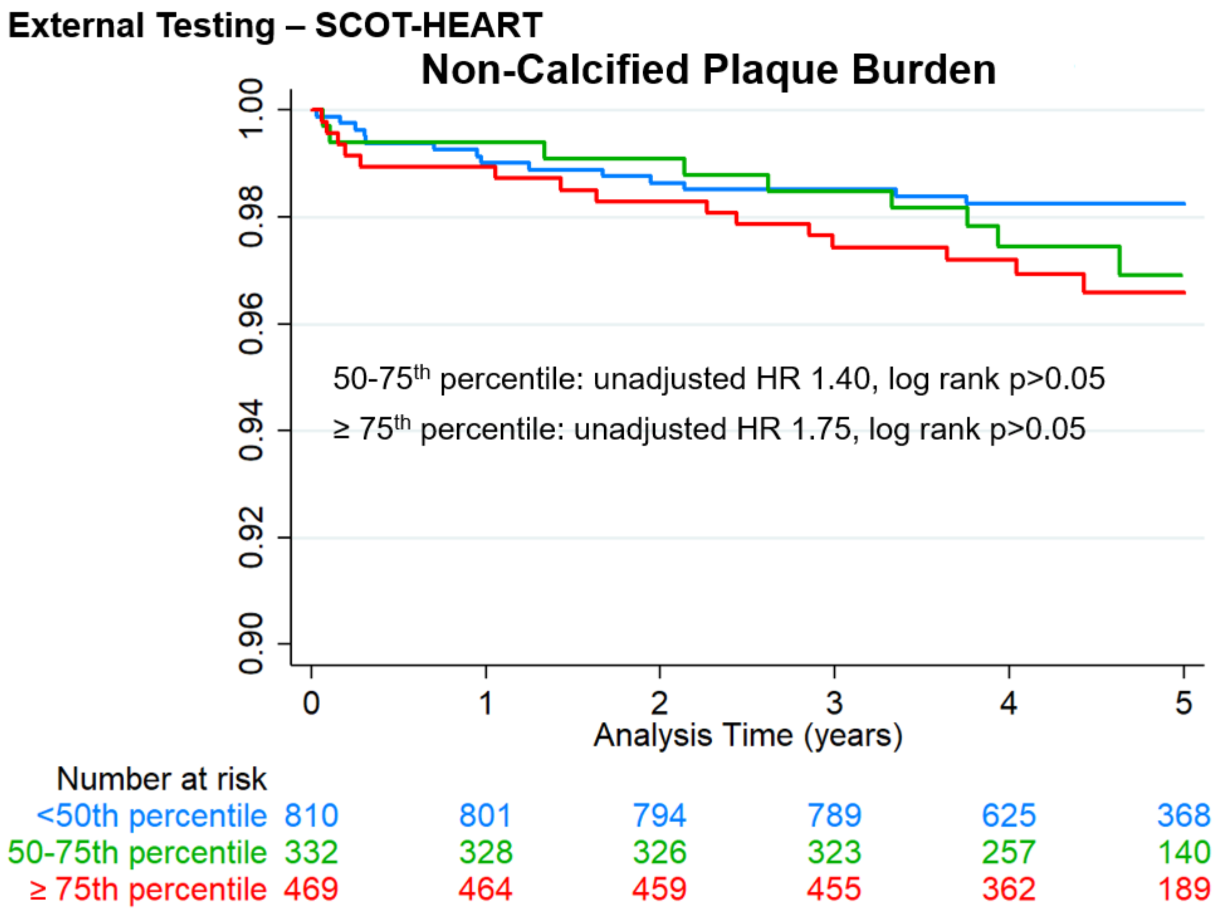
Survival free of fatal or non-fatal myocardial infarction stratified by percentile of low-attenuation plaque volume. Reference population was patients in the <50[th] percentile for age and sex. The percentiles were derived in a separate population, then applied to a cohort of patients from the Scottish Computed Tomography of the Heart trial. HR – hazard ratio

Figure S10



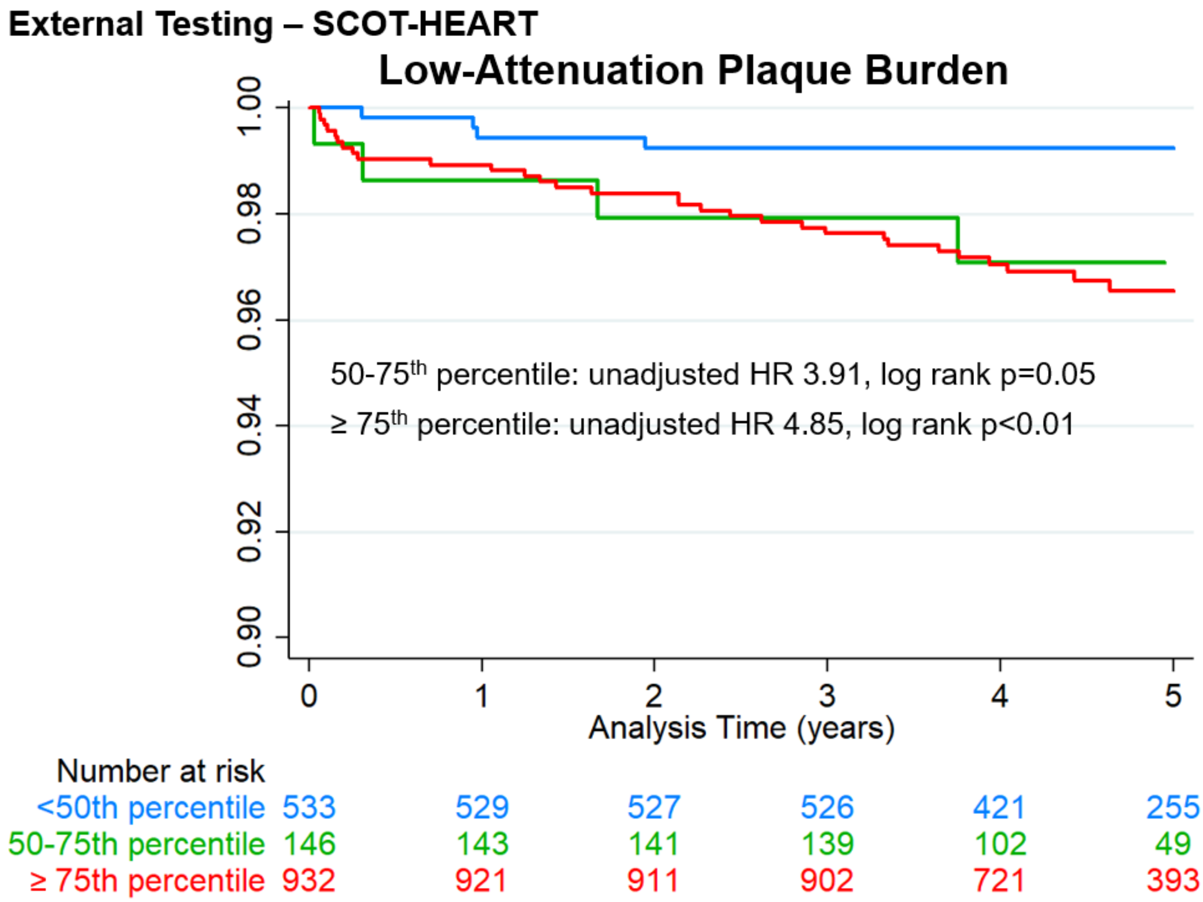
Survival free of fatal or non-fatal myocardial infarction stratified by percentile of calcified plaque volume. Reference population was <50th percentile for age and sex. The percentiles were derived in a separate population, then applied to a cohort of patients from the Scottish Computed Tomography of the Heart trial. HR – hazard ratio

Figure S11



Survival free of fatal or non-fatal myocardial infarction stratified by percentile of non-calcified plaque burden. Reference population was <50th percentile for age and sex. Plaque burden is calculated as plaque volume/analyzed segment volume. The percentiles were derived in a separate population, then applied to a cohort of patients from the Scottish Computed Tomography of the Heart trial. HR – hazard ratio

Figure S12



Survival free of fatal or non-fatal myocardial infarction stratified by percentile of low-attenuation plaque burden. Reference population was <50th percentile for age and sex. Plaque burden is calculated as plaque volume/analyzed segment volume. The percentiles were derived in a separate population, then applied to a cohort of patients from the Scottish Computed Tomography of the Heart trial. HR – hazard ratio

Table S1

Cohort	N	Mean Age	% Male	Details
Cedars-Sinai, Symptomatic	223	66.0 ± 10.8	81.2	Registry of patients undergoing clinically indicated coronary CTA with symptoms of CAD
Cedars-Sinai, Asymptomatic	388	59.8 ± 11.7	68.6	Registry of patients undergoing clinically indicated coronary CTA without symptoms of CAD
Diamond (NCT02110303)	148	65.2 ± 8.0	85.8	Single-center randomized trial of antiplatelet therapy in patients with multivessel, stable CAD. Baseline CTA before randomization.
Kusatsu	72	67.2 ± 10.4	81.9	Registry of patients with known or suspected CAD who underwent IVUS within 30 days of coronary CTA. Excluded patients with acute myocardial infarction or stented lesions.
Oklahoma	125	61.0 ± 10.9	59.2	Patients with symptoms of CAD undergoing clinically indicated coronary CTA

Details of the derivation cohorts. The cohorts included both symptomatic and asymptomatic patients are described. CAD – coronary artery disease, CTA – computed tomography angiography, IVUS – intravascular ultrasound.

Table S2

Female Patients:

	Age ≤45	Age 46-55	Age 56-65	Age 66-75	Age >75
<10th percentile	8	18	22	18	5
10-25th percentile	0	0	0	0	0
25-50th percentile	0	0	0	11	10
50-75th percentile	0	0	29	38	20
75-90th percentile	1	10	15	15	9
90-95th percentile	2	1	3	5	0
≥95th percentile	0	1	5	3	0

Male Patients:

	Age ≤45	Age 46-55	Age 56-65	Age 66-75	Age >75
<10th percentile	26	39	22	39	17
10-25th percentile	0	0	26	26	10
25-50th percentile	0	22	48	27	4
50-75th percentile	15	29	53	58	20
75-90th percentile	8	30	43	37	9
90-95th percentile	3	8	16	14	5
≥95th percentile	1	11	12	23	6

Distribution of number of patients according to percentiles of total plaque volume stratified by age and sex. All patients with plaque volumes of 0 are categorized as 0th percentile.

Table S3

	Clinical and CAC model		Stenosis Model		Plaque Model	
Variable	Adjusted HR (95% CI)	p-value	Adjusted HR (95% CI)	p-value	Adjusted HR (95% CI)	p-value
Age (per year)	1.01 (0.99 – 1.04)	0.295	1.01 (0.99 - 1.04)	0.386	1.01 (0.98 - 1.04)	0.409
Male	1.24 (0.78 – 1.99)	0.362	1.21 (0.75 - 1.97)	0.435	1.21 (0.72 - 2.03)	0.468
Hypertension	0.99 (0.61 – 1.61)	0.980	0.99 (0.61 - 1.61)	0.982	0.99 (0.61 - 1.61)	0.981
Diabetes	1.21 (0.66 – 2.22)	0.541	1.21 (0.66 - 2.22)	0.540	1.21 (0.66 - 2.22)	0.541
Dyslipidemia	1.00 (0.64 – 1.57)	0.988	1.00 (0.64 - 1.56)	0.999	1 (0.64 - 1.56)	1
Smoking	0.96 (0.52 – 1.77)	0.886	0.95 (0.51 - 1.76)	0.863	0.95 (0.51 - 1.77)	0.866
Family History	1.06 (0.67 – 1.68)	0.805	1.07 (0.67 - 1.7)	0.785	1.07 (0.67 - 1.7)	0.784
CAC Score (per 100)	1.02 (0.99 – 1.04)	0.200	1.02 (0.99 - 1.04)	0.233	1 (1 - 1)	0.39
Obstructive CAD	--	--	1.12 (0.64 - 1.96)	0.692	1.12 (0.63 - 1.98)	0.705
TP volume (per mm^3)	--	--	--	--	1 (1 - 1)	0.976
TP volume Percentile						
< 50th	Reference	--	Reference	--	Reference	--
50-75th	**2.22 (1.32 – 3.74)**	**0.003**	**2.17 (1.27 - 3.7)**	**0.005**	**2.16 (1.25 - 3.73)**	**0.005**
≥ 75th	**2.50 (1.29 – 4.86)**	**0.007**	**2.39 (1.19 - 4.83)**	**0.015**	**2.38 (1.06 - 5.33)**	**0.035**
Harrell's C without TP percentile	0.617		0.595		0.595	
Harrell's C with TP percentile	0.623		0.612		0.611	

Results for the multivariable analyses considering total plaque (TP) volume percentiles. Analyses performed in the combined external sample. Significant values in bold. CAC – coronary artery calcium, CI – confidence interval, HR – hazard ratio.

Table S4

	Clinical and CAC model		Stenosis Model		Plaque Model	
	Categorical NRI (95% CI)	p-value	Categorical NRI (95% CI)	p-value	Categorical NRI (95% CI)	p-value
Total plaque volume	**0.205 (0.120 to 0.289)**	**<0.001**	**0.259 (0.173 to 0.344)**	**<0.001**	**0.265 (0.177 to 0.352)**	**<0.001**
Non-calcified plaque volume	**0.206 (0.119 to 0.293)**	**<0.001**	**0.216 (0.139 to 0.292)**	**<0.001**	**0.186 (0.112 to 0.259)**	**<0.001**
Low-attenuation plaque volume	**0.216 (0.120 to 0.311)**	**<0.001**	**0.264 (0.172 to 0.355)**	**<0.001**	**0.159 (0.097 to 0.221)**	**<0.001**
Calcified plaque volume	**0.107 (0.030 to 0.184)**	**0.006**	**0.181 (0.116 to 0.245)**	**<0.001**	**0.243 (0.151 to 0.335)**	**<0.001**
Non-calcified plaque burden	**0.084 (0.006 to 0.163)**	**0.036**	**0.119 (0.064 to 0.174)**	**<0.001**	-0.039 (-0.078 to 0.002)	0.056
Low-attenuation plaque burden	**0.189 (0.099 to 0.279)**	**<0.001**	**0.227 (0.1434 to 0.310)**	**<0.001**	0.056 (-0.004 to 0.122)	0.065

Results of net-reclassification index (NRI) analysis. The clinical and coronary artery calcium (CAC) model included age, sex, risk factors (hypertension, diabetes, dyslipidemia, smoking, family history), and CAC score. The stenosis model included the variables from the clinical and CAC model variables as well as obstructive coronary artery disease (defined as left main stenosis ≥50% or ≥70% in other segments). The plaque model included the variables from the stenosis model as well as plaque volume or burden as a continuous variable. Categories were very low (<1%), low (1-2%), intermediate (2-5%) and high (>5%) risk of events during follow-up. Significant values in bold. CI – confidence interval.